

# SemanticDistance: An R package for computing semantic relationships in language samples

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## Summary

Word meanings can be represented as vectors in n-dimensional semantic space. The distance between successive items in a list, sentence, or story is important for understanding the type of information being conveyed. **SemanticDistance** is an R package for flexibly estimating the distance between words or groups of words in a text.

## Statment of Need

One of the specific things listeners do in the process of understanding a conversation or story is to relate incoming words with what has been previously heard. *Semantic distance* (Reilly et al., 2023) corresponds to the dissimilarity between two or more concepts within an n-dimensional space, typically derived from analyzing word co-occurrence in large corpora of texts (Landauer & Dumais, 1997). Currently there are no existing packages that implement these calculations.

## Description

The **SemanticDistance** R package is available from:

<https://github.com/Reilly-ConceptsCognitionLab/SemanticDistance>

Consider, for example, a researcher interested in quantifying the distance between *wolf* and *dog* in a unidimensional semantic space constrained by perceived threat. A simple subtraction of the respective threat ratings for *wolf* and *dog* would yield an empirical index of the distance between these two concepts in ‘threat’ space. In practice, most researchers interested in modeling semantic relationships do so using multidimensional semantic spaces. This approach involves quantifying the salience of target words across many unique psychological dimensions (e.g., color, sound, threat, etc.) or in the case of word embedding models across a series of hyperparameters.

**SemanticDistance** appenda distance values between each pair of elements specified by the user (e.g., word-to-word, ngram-to-word, ngram-to-ngram). These distance values are derived from two large lookup databases in the package with fixed semantic vectors for >70k English words. **CosDist\_Glo** reflects cosine distance between vectors derived from training a GLOVE word embedding model (300 hyperparameters per word) (Pennington

et al., 2014). `CodDist_SD15` reflects cosine distance between two chunks (words, groups of words) characterized across 15 meaningful perceptual and affective dimensions (e.g., color, sound, valence).

Users specify an ngram window size. This window rolls successively over a language sample to compute a semantic distance value for each new word relative to the n-words (ngram size) before it. A 1-gram distance computes the distance from word-to-word; a 2-gram would compute the distance from a pair of words to the next pair, and so on.

## Preparation of text

Before using `SemanticDistance`, users need to decide what format their text is in and what they want to measure. `SemanticDistance` offers many possible options with some default arguments. For example, the package requires users to clean and prepare the data. Users can choose to omit stopwords, lemmatize, split strings, and so on. Or, users can decide to leave their data alone and split the transcript into a one-word-per-row format. The prepared dataframe should minimally contain a text column and a speaker/talker column.

```
library(SemanticDistance)

Monologue_Cleaned <- clean_monologue(dat=Monologue_Structured,
                                     wordcol='mytext', clean=TRUE,
                                     omit_stops=TRUE, split_strings=TRUE)

head(Monologue_Cleaned, n=8)
```

```
## # A tibble: 8 x 5
##   id_row_orig word_clean timestamp mytext   id_row_postsplit
##   <fct>      <chr>          <int> <chr>          <int>
## 1 1          <NA>            1 "the"           1
## 2 2          girl            2 "girl"          2
## 3 3          walk            3 "walked"        3
## 4 4          down            4 "down "         4
## 5 5          <NA>            5 "the "          5
## 6 6          street          6 "street"        6
## 7 7          <NA>            7 "the"           7
## 8 8          boxer            8 "boxer"         8
```

## Semantic distance estimates

Included function average the semantic vectors for all content words in a turn then computes the distance to the average of the semantic vectors of the content words in the subsequent turn. It averages across the semantic vectors of all words within a turn and then computes cosine distance to all the words in the next turn. A user simply needs to feed it a transcript formatted with `clean_dialogue`. The function `dist_dialogue` returns a summary dataframe that distance values aggregated by talker and turn (`id_turn`).

```
Ngram2Ngram_Dist1 <- dist_ngram2ngram(dat=Monologue_Cleaned, ngram=2)
head(Ngram2Ngram_Dist1)
```

```
## # A tibble: 6 x 7
##   id_row_orig word_clean timestamp mytext   id_row_postsplit CosDist_2gram_GLO
##   <fct>      <chr>          <int> <chr>          <int>          <dbl>
## 1 1          <NA>            1 "the"           1             NA
## 2 2          girl            2 "girl"          2             NA
## 3 3          walk            3 "walked"        3             NA
## 4 4          down            4 "down "         4             0.141
```

```
## 5 5          <NA>          5 "the "          5          0.0608
## 6 6          street        6 "street"         6          0.319
## # i 1 more variable: CosDist_2gram_SD15 <dbl>
```

## Visualization

`SemanticDistance` allows several visualizations of the data. These include cluster and dendrogram visualizations of how words in a text sample relate to one another.

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## References

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